# The Pareto Landscape of Moral Reasoning

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## Abstract

Evaluating moral reasoning is hard because the moral domain lacks the ground truth that organises evaluation elsewhere: competent reasoners can start from similar intuitions and reach different defensible conclusions. We argue that this is not a measurement failure but a structural feature, and we propose a conceptual framework that takes it seriously. Moral reasoning, on our account, aims at a *good epistemic state*: a pair of commitments and principles that hangs together. Such a state is graded along two axis families that genuinely conflict: epistemic desiderata (account, systematicity, faithfulness) and value loadings (freedom, well-being, justice, dignity, solidarity). The rational object of study is then the *Pareto frontier* of attainable epistemic states. We develop a vocabulary for the shapes this frontier takes (convex segments, concave segments, knees, plateaus, corner attractors, dominated regions), and we recast familiar reasoning methods, reflective equilibrium foremost, as strategies for navigating the landscape toward the frontier. We sketch the corpus that would let this framework be studied empirically, and outline what such study would reveal about moral reasoning in humans and in language models.

## 1 Motivation

Current alignment work focuses on the moral *content* a model holds: constitutional principles, character training, value specifications. Less attention has gone to moral reasoning as a *capacity* (Carlsmith, 2026), and existing evaluations tend to conflate the two (Haas et al., 2026). But a value-aligned model whose reasoning fails when its values come into tension is still unsafe: it can produce outputs that look like principled moral thought without being so. Diagnosing such failures is hard because the moral domain lacks a ground truth against which to score answers (Queloz, 2025): competent reasoners can reach different conclusions without either being simply wrong.

This paper proposes a framework that takes the absence of ground truth as a *structural feature* rather than a measurement failure. We articulate what moral reasoning aims at, what makes its outputs better or worse without a single right answer, and how the multi-objective optimisation literature (Deb, 2001) gives us a precise vocabulary for the kinds of trade-offs the moral domain actually exhibits. Familiar methods such as reflective equilibrium (Rawls, 1971; Knight, 2023), recently formalised by Beisbart et al. (2021) and studied at simulation scale by Freivogel (2023), reappear as *strategies for navigating* this landscape, comparable in principle to alternative strategies and amenable to empirical study once a suitable corpus is in place.

## 2 Framework

We develop the framework in four steps: introducing the conceptual objects moral reasoning operates on, the axes along which their combinations are graded, the Pareto frontier those axes define, and a vocabulary for the shapes the frontier can take.

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### 2.1 Commitments, principles, epistemic state

An **ethical situation** confronts an agent with a decision: several actions are open, and none is clearly right or wrong. On encountering such a situation the agent has **commitments**: pre-theoretical verdicts about how to act in the case, of the form "in situation $s$, action $a$ is required / permissible / forbidden." Across the body of situations the agent has considered, these commitments form a set $C$. A reasoner holding $C$ alone has no resources to handle a new situation, to defend a verdict against an objection, or to notice when two commitments pull in opposite directions.

This is where **principles** enter. A principle $p$ is a general moral rule, independent of any particular situation, that an agent endorses as justifying some of their commitments. The agent's principles together form a set $P$. Principles do three kinds of work simultaneously: they *account* for commitments (giving each commitment a reason), they *generalise* to situations the agent has not yet considered (a single principle yields verdicts across many cases), and they *regulate* the commitments (revealing inconsistency when two commitments cannot both follow from the same principle).

The output of moral reasoning is an **epistemic state** $\sigma = (C, P)$: a set of commitments paired with a set of principles that hangs together. We take the goal of moral reasoning to be reaching a *good* epistemic state. Two families of demand pull on what *good* means here. The state should satisfy the agent's *epistemic* expectations of reasoning: the principles should account for the commitments, they should be simple and broadly applicable rather than a hodgepodge of one-offs, and the commitments should not drift arbitrarily from the agent's starting intuitions $C_0$. The state should also fit the agent's *ethical* profile: the principles should express the values and deeper ethical frameworks the agent endorses on reflection, and where values trade off, the trade-off the state strikes should be one the agent can stand behind. We make these two families precise in Section 2.2.

### 2.2 What makes an epistemic state good

Two axis families grade $\sigma$ simultaneously. They genuinely conflict, in the sense that improving one typically costs another.

**Epistemic axes.** Properties of $\sigma$ as reasoning. Following the formal account of Beisbart et al. (2021), we work with three; others could be envisaged (simplicity, scope, robustness) and the framework accommodates them without modification.

- ACCOUNT: how much the principles in $P$ entail or support the commitments in $C$, penalising contradictions, gaps, and surplus content.
- SYSTEMATICITY: how simple, unifying, and broadly applicable $P$ is, as opposed to a hodgepodge of case-by-case stipulations.
- FAITHFULNESS: how well the current commitments respect the agent's initial intuitions $C_0$, penalising arbitrary drift.

**Ethical axes.** Properties of $\sigma$ as a moral stance. Each principle $p \in P$ carries a fuzzy *value loading* $\boldsymbol{v}(p) \in [0,1]^K$ over a small set of basic moral values. We work with five: FREEDOM, WELL-BEING, JUSTICE, DIGNITY, and SOLIDARITY, chosen so that the central tensions of moral life (liberty vs. community, fairness vs. welfare, welfare vs. dignity) appear directly as trade-offs among axes. The state $\sigma$ inherits an aggregate value position by aggregating its principles' loadings: the more a value is activated by the principles in $P$, the more the state weighs on that axis. Concretely, $V(\sigma) = \frac{1}{|P|}\sum_{p \in P} \boldsymbol{v}(p) \in [0,1]^K$, so locating an epistemic state in the ethical-value subspace amounts to checking which values each of its principles activates.

**Two families, two axes of disagreement.** The epistemic family is broadly agent-independent: every reasoner has some stake in coherence. The ethical family is agent-dependent: different agents weight different values, and this pluralism is a constitutive feature of the moral domain rather than noise to be averaged out.

### 2.3 Reasoning moves

Reaching a good $\sigma$ involves recognisable operations on $C$ and $P$. We propose a candidate list of typed primitives, each with an input-output signature that makes it addressable as a unit of evaluation independent of the strategy that composes it. The first four are needed, to varying degrees, by any moral reasoning strategy; the last two are characteristic of *dialectical* strategies and will reappear in Section 3 as the central moves of reflective equilibrium.

- **Property elicitation** $s \to \{\varphi_1, \dots, \varphi_m\}$. From a situation, make explicit the ethically relevant properties: who is affected, what interests are at stake, the kind of action on offer. The basis on which a commitment will be formed.
- **Principle proposal** (abduction) $\{c_1, \dots, c_n\} \to p$. From a set of commitments, propose a candidate principle that justifies them. The inductive move from cases to rule.
- **Principle deduction** $(p, s) \to c$. From a candidate principle and a situation, derive the commitment the principle implies. The downward complement of proposal.
- **Coherence checking** $\sigma \to \mathbb{R}^{3+K}$. Score the current state on the epistemic and ethical axes (Section 2.2). A meta-operation: it does not move $\sigma$, but it tells any other operation whether its candidate move improves things.
- **Commitment revision** $(c, p) \to c'$. Adjust a commitment in light of a principle that conflicts with it.
- **Principle revision** $(p, c) \to p'$. Adjust a principle in light of a commitment that conflicts with it.

The dialectical alternation of the two revisions (Rawls, 1971; Beisbart et al., 2021) is what gives reflective equilibrium its name; we make the strategy explicit in Section 3, after spelling out the space the moves take place in.

### 2.4 The Pareto frontier and its projections

We can now state the framework precisely. Let $\Sigma$ be the set of all internally consistent epistemic states. Each $\sigma \in \Sigma$ has a position in the joint objective space

$$\big(A(\sigma),\, S(\sigma),\, F(\sigma \mid C_0),\, V_1(\sigma), \dots, V_K(\sigma)\big) \in [0,1]^{3+K}.$$

Multi-objective optimisation (Deb, 2001) gives us the right vocabulary for the structure of this space when its axes genuinely conflict.

**Pareto dominance.** A state $\sigma'$ *Pareto dominates* $\sigma$, written $\sigma \prec \sigma'$, if every objective at $\sigma'$ is at least as large as at $\sigma$ and at least one is strictly larger.

**The Pareto frontier.** The frontier $\Pi$ is the set of non-dominated states in $\Sigma$:

$$\Pi \;=\; \{\, \sigma \in \Sigma : \nexists\, \sigma' \in \Sigma \text{ with } \sigma \prec \sigma' \,\}.$$

$\Pi$ is what we mean by *the space of defensible epistemic states*. For any state off the frontier, some defensible move strictly improves it on every axis the agent cares about; for any state on the frontier, no such move exists.

**Untangling $\Pi$ into two projected frontiers.** $\Pi$ lives in the joint $(3 + K)$-dimensional space and treats all axes on equal footing. For analysis it is often convenient to *untangle* $\Pi$ into its two natural projections: the *epistemic Pareto frontier* $\Pi_E$, the set of non-dominated states in the $(A, S, F)$ subspace, and the *ethical Pareto frontier* $\Pi_V$, the set of non-dominated states in the $(V_1, \dots, V_K)$ subspace (Figure 1). The two need not coincide: a state can sit on $\Pi_E$ (epistemically coherent) without sitting on $\Pi_V$ (ethically Pareto-optimal), and vice versa. A good epistemic state lies on both.

**Pluralism is not subjectivism.** Although the choice of a point on $\Pi_V$ is agent-dependent (different value profiles favour different points), the frontier itself is not: the set of defensible states can in principle be characterised independently of any value profile. This suggests a productive division of labour for an ideal artificial moral reasoner: map the full frontier and present its structure, leaving the deliberative choice of point to human reasoners.

### 2.5 Vocabulary for the shape of the frontier

Different ethical situations, and sets of situations grouped into a context or domain, produce Pareto frontiers of different shapes. The multi-objective optimisation literature already has a rich vocabulary for the shapes these frontiers can take, which we can borrow directly.

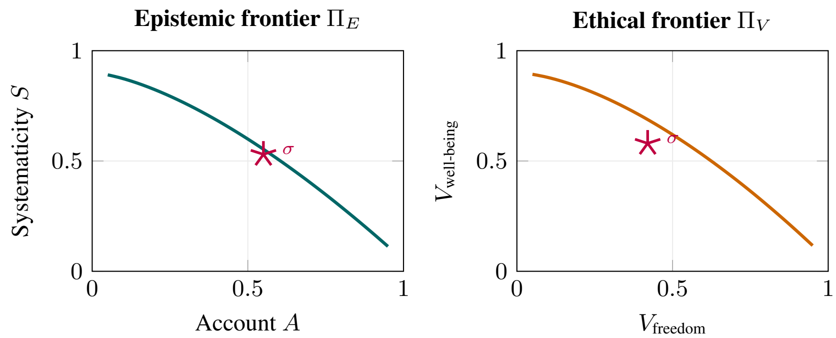


Figure 1: **The two projected Pareto frontiers (schematic).** Untangling the joint frontier $\Pi$ yields the epistemic frontier $\Pi_E$ (left) and the ethical frontier $\Pi_V$ (right). A state $\sigma$ has coordinates in both subspaces. The two projected frontiers need not coincide; a good $\sigma$ lies on both.

- **Convex segments.** Weighted sums of axes pick a unique answer. There is little genuine disagreement: reasoners differ in weights but agree in conclusion.
- **Concave segments.** Incomparable defensible answers: no weighting of axes selects between them. *This is where genuine pluralism lives.*
- **Knee points.** A small concession on one axis buys a large gain on another. The *natural compromise* of the case, where reasoning tends to land when neither value dominates.
- **Plateaus.** Many near-equivalent positions. The choice is under-determined and competent reasoners may land in slightly different places without error.
- **Corner attractors.** Extreme positions where one axis dominates and others collapse. Pure utility, pure autonomy, pure procedural justice. Often degenerate.
- **Dominated region.** Bad reasoning. Off-frontier states are off-frontier precisely because some defensible move strictly improves them.

Figure 2 illustrates each shape on a stylised two-axis front.

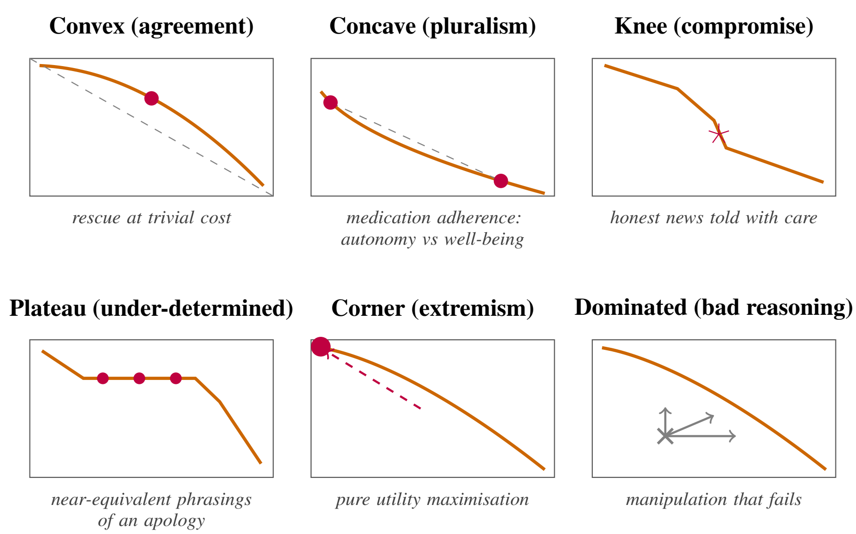


Figure 2: **Shapes the Pareto frontier can take (schematic).** Each panel pairs one shape with a moral situation that would typically exhibit it. Stylised, not derived from data.

**An illustrative example.** Consider a user who refuses a prescribed medication, partly because of delusional thinking, while interacting with an LLM acting as a therapeutic interlocutor. The values primarily in tension are FREEDOM and WELL-BEING. Two responses sit on a concave segment of Π: an autonomy-leaning one states the medical facts plainly and does not push further, and a care-leaning one explores the refusal through dialogue and non-manipulative persuasion. Neither dominates the other. An interior knee point may exist (a specific autonomy-preserving care strategy that buys large WELL-BEING gain for small FREEDOM cost). Several responses are off-frontier and dominated: insulting the user, manipulating without succeeding, walking away without escalation. Figure 3 visualises this landscape.

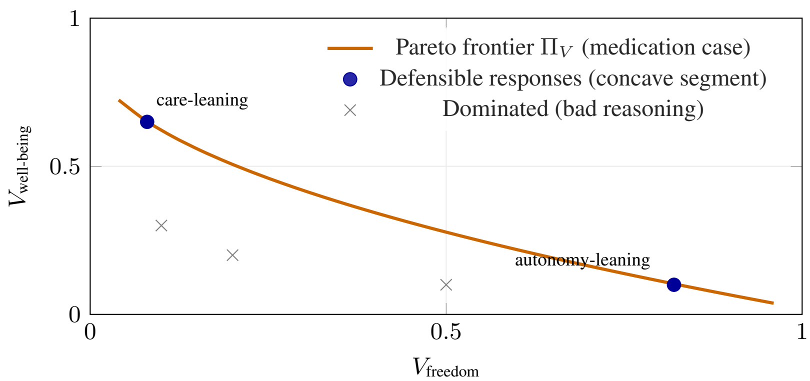


Figure 3: Sketch of the ethical-value Pareto frontier in the medication case, projected onto FREEDOM and WELL-BEING. The frontier is concave between the two defensible responses: weighted sums of the two axes favour the endpoints, and the segment between them is the locus of genuine pluralism. Several reachable responses sit strictly inside the dominated region.

## 3 Strategies for navigating the landscape

If moral reasoning aims at the frontier, the natural follow-up question is *how an agent gets there*. A strategy produces a *reasoning trace*: a finite sequence $\sigma_0, \sigma_1, \ldots, \sigma_n$ of epistemic states linked by the typed operations of Section 2.3. Each move has a characteristic contribution to the axes: principle proposal typically lifts ACCOUNT and SYSTEMATICITY together; commitment revision trades FAITHFULNESS for ACCOUNT; principle revision trades SYSTEMATICITY for ACCOUNT, or one value for another. Figure 4 shows three strategies as paths from a shared initial commitment toward Π.

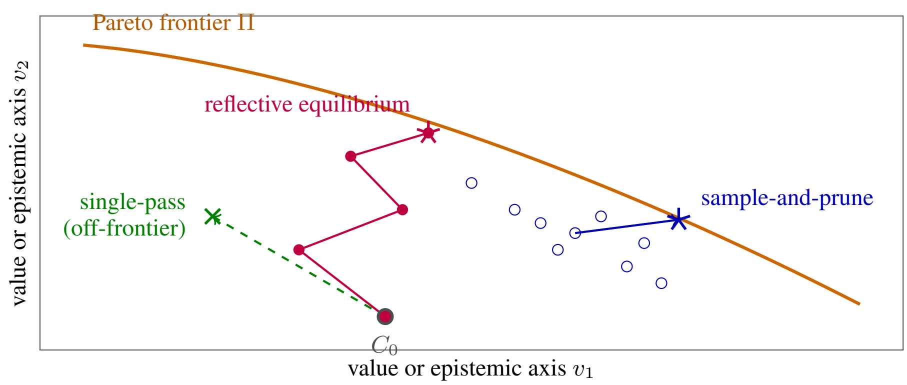


Figure 4: **Reasoning as movement toward the Pareto frontier (schematic).** Reflective equilibrium walks dialectically from the initial commitments $C_0$ and ends at a frontier point. Sample-and-prune explores broadly and selects the non-dominated candidate. Single-pass abduction leaps once and may land in the dominated region. Shapes and trajectories are stylised, not derived from data.

### 3.1 Reflective equilibrium

Reflective equilibrium is widely regarded by philosophers as the canonical methodological framework for moral reasoning (Rawls, 1971; Knight, 2023): a good moral position is one the agent reaches by mutually adjusting commitments and principles until the two hang together. Concretely, begin with $\sigma_0 = (C_0, P_0)$, where $C_0$ is the agent's initial commitments and $P_0$ is some starting theory (possibly empty). The procedure alternates two kinds of revision, scoring the result at each step:

(1) *Principle revision* $(p, c) \to p'$. Holding $C$ fixed, propose a revised theory that better accounts for the current commitments. Score the resulting state via coherence checking; accept if it improves.

(2) *Commitment revision* $(c, p) \to c'$. Holding $P$ fixed, propose a revised commitment set that better aligns with the current theory. Score the resulting state; accept if it improves.

Alternate the two steps until neither produces an improvement. The fixed point $\sigma^\star$ is the equilibrium. The accept rule at each step is itself a design choice; Beisbart et al. (2021) use the weighted scalar aggregate $Z = \alpha_A A + \alpha_S S + \alpha_F F$. Reflective equilibrium is well-suited to finite reasoners with strong intuitions and a limited search budget: each step is local, no global search is required, and the procedure terminates when intuitions and principles are mutually adjusted.

### 3.2 Alternative strategies

Reflective equilibrium is one strategy among several. Plausible alternatives include:

- *Sample-and-prune*: generate many candidate $(C, P)$ configurations from the pools, score each, retain the non-dominated ones. A direct attempt to map a region of $\Pi$ rather than walk toward a single point. Well-suited to reasoners with cheap sampling.
- *Tree search*: build a tree of reasoning moves from $\sigma_0$, expand promising branches, backtrack from dead ends. A natural fit for artificial reasoners with a notion of branch quality and a backtracking mechanism.
- *Single-pass abduction*: propose one principle from $C_0$, deduce its consequences, accept the result. The naive baseline, often under-evaluated against the others.
- *Multi-start equilibrium*: run reflective equilibrium from several distinct starting points to discover multiple regions of $\Pi$ at once. Particularly useful when the frontier is expected to be concave or plateau-shaped.
- *Corner-collapse*: maximise a single axis (pure ACCOUNT, pure WELFARE, pure AUTONOMY). Degenerate, but instructive as an ablation and as a failure mode of stronger strategies.

### 3.3 Which strategy for which reasoner?

For humans, reflective equilibrium is the canonical answer, in part because human deliberation has a budget that favours local moves over wide sampling. For artificial reasoners (LLMs in particular), the question is open. Sampling-based strategies may exploit parallelism that humans cannot; tree-search strategies may map richer landscapes; the failure modes of single-pass strategies may or may not appear under chain-of-thought. The framework lets us ask these questions on equal footing once we can locate any strategy's output in the space.

## 4 Empirical agenda

The framework so far is conceptual. We sketch the corpus that would let it be studied empirically, and the questions that corpus would make tractable.

### 4.1 A corpus to make this empirical

We sketch what we would dream of having and what it would let us do.

**Contents.** For a body of ethical situations, the corpus would contain three layers.

1. **Commitment pool** $\mathcal{C}$: candidate verdicts on each situation. Per-case action judgments of the form "in situation $s$, action $a$ is required / permissible / forbidden," generated broadly enough to cover the defensible verdicts plus a margin of bad ones.
2. **Principle pool** $\mathcal{P}$: candidate moral principles, each tagged with a *fuzzy value loading* $\boldsymbol{v}(p) \in [0,1]^K$ indicating to what degree the principle expresses each value. Optionally, a secondary tagging by ethical framework $\boldsymbol{f}(p) \in [0,1]^M$ (utilitarian, Kantian, virtue, care, contractualist) supports framework-level analyses without committing to frameworks as primary axes.
3. **Dialectical graph** $\rho$: for every principle-commitment pair, a *fuzzy inferential strength* $\rho(p,c) \in [-1,1]$, where positive values indicate degree of entailment, negative values degree of contradiction, and zero silence. This generalises a discrete entails / contradicts / silent labelling and captures the partial support relations that are typical in moral reasoning.

**Construction.** The corpus would be generated by a large language model and verified by philosophers at every stage: generation, fusion of near-duplicates, deletion of weak candidates, scoring of independent quality dimensions, value tagging, and inferential strengths. We treat the corpus as a reusable resource in its own right, independent of any downstream reasoning method that operates on it.

**What it would let us do.** Given a starting set of intuitions $C_0$ drawn from $\mathcal{C}$, the corpus lets us:

- Compute the Pareto fronts $\Pi_E$ and $\Pi_V$ of attainable epistemic states, and characterise their shapes locally and globally.
- Locate any epistemic state $\sigma$ in the joint space, and say whether it is on $\Pi$, near $\Pi$, or dominated, and if dominated, by what.
- Inspect reasoning traces move by move: is each step principled (a justified abduction, a defensible commitment revision) or unprincipled (a faithfulness violation with no compensating gain)? Does the trajectory walk toward $\Pi$ or away from it?
- Compare reasoning strategies on equal footing.
- *Evaluate each typed primitive of Section 2.3 in isolation.* Given the dialectical graph and the value loadings, we can score how well a model proposes principles from a held-out set of commitments, deduces verdicts from a given principle, revises a commitment under a conflicting principle, and so on, without committing to any particular composition of these primitives.

### 4.2 What we can study

With the framework and a corpus in place, three families of empirical question become tractable: questions about the moral domain itself, questions about how reasoning strategies behave, and questions about how language models in particular reason morally.

**The geometry of the moral domain.** For a given case, is the local shape of $\Pi$ convex (resolvable disagreement) or concave (genuine pluralism)? Does the case admit a knee point (a natural compromise)? Across many cases, what shapes recur, and what does that tell us about the structure of the moral domain itself?

**Comparing reasoning strategies.** Do different strategies reach different regions of $\Pi$ on the same case? Does reflective equilibrium land in plateaus while sample-and-prune maps concave segments? Which strategy is more robust under reframing of the case? The corpus puts strategies on equal footing, replacing incommensurable benchmarks with a common measurable space.

**LLM failure modes at the trace level.** A value-aligned model whose traces sit off $\Pi$ in characteristic ways is a value-aligned model that reasons badly, and the framework lets us name how: *corner collapse* (the trace ends at an axis extreme), *anti-revision rigidity* (no move revises the principle side), *faithfulness violations* (commitments dropped with no justifying gain), *stability collapse* (different landings under semantically equivalent rephrasings of the situation). Each becomes an empirically testable hypothesis about LLM behaviour once the framework names it.

**LLM capacities at the primitive level.** The typed operations of Section 2.3 are candidate irreducible capacities of moral reasoning. The corpus lets us probe each in isolation: which capacities does the model have, and at what quality? Per-primitive diagnostics give a finer-grained picture than trace-level analyses, and may reveal that a model competent at proposing principles is weak at revising them under counterevidence, or vice versa.

**Designing better artificial moral reasoners.** The framework also supports a constructive agenda. If a strategy turns out to reach $\Pi$ reliably and to land in defensible regions of it across many cases, it becomes a natural training target. Reflective equilibrium is the obvious candidate, given its philosophical pedigree, but the framework lets the comparison be empirical rather than dogmatic. The strategy that wins can then be operationalised: a system that explicitly composes the typed primitives in the strategy's order, or a model fine-tuned on traces produced by the strategy applied to a wide range of cases. Evaluation and construction become two sides of the same framework.

## 5 Discussion

**Related work.** Our framework draws on three strands that have not previously been brought together. The first is the formal model of reflective equilibrium developed by Beisbart et al. (2021) and the simulation study of Freivogel (2023), which we adopt for the epistemic desiderata. The second is the literature on moral pluralism and the asystematicity of normative domains (Queloz, 2025; Elgin, 1996), which motivates treating the absence of ground truth as structural. The third is multi-objective optimisation (Deb, 2001), from which we borrow the vocabulary of dominance, frontier, and shape. The empirical agenda our framework supports is most closely related to recent work on evaluating moral reasoning in language models (Haas et al., 2026; Awad et al., 2022), with the difference that our unit of analysis is the *trace* and its location in a measurable space rather than aggregate behaviour on a benchmark.

**Outlook.** The framework gives the familiar pieces of moral reasoning a clearer place: the goal is a good epistemic state, goodness is graded along epistemic and ethical axes that genuinely conflict, the rational object of study is the Pareto frontier of attainable states, and methods such as reflective equilibrium are strategies for navigating it. What remains is the corpus (Section 4.1) and the empirical study it supports (Section 4.2). Both are non-trivial to build, but both are now well-posed: the framework tells us what to measure, what shapes to look for, and what failure modes to name.

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